



Group by trace, then solve for S

1 Concrete terms



$$I \otimes I$$



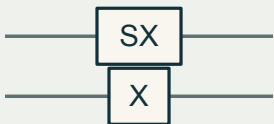
$$CZ$$



$$CX$$



$$X \otimes I$$

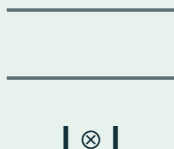


$$SX \otimes X$$

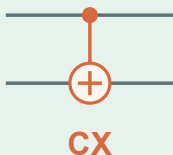
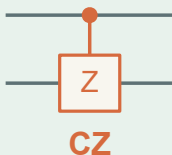
2 qubits · 10 pairs

2 Group by concrete trace

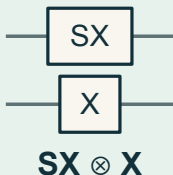
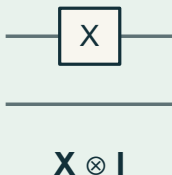
tr = 4



→ **tr = 2**



tr = 0



2 pairs survive

3 Solve $SL = RS$

$$SL = RS$$

$$(L, R) = (CZ, CX)$$


$$S = \begin{bmatrix} a & b & c & 0 \\ d & e & f & 0 \\ g & h & j & k \\ g & h & j & -k \end{bmatrix}$$

$$(L, R) = (X \otimes I, SX \otimes X)$$

$$S = \begin{bmatrix} a & b & c & d \\ c & d & a & b \\ a & b & c & d \\ c & d & a & b \end{bmatrix}$$

10-dimensional linear space

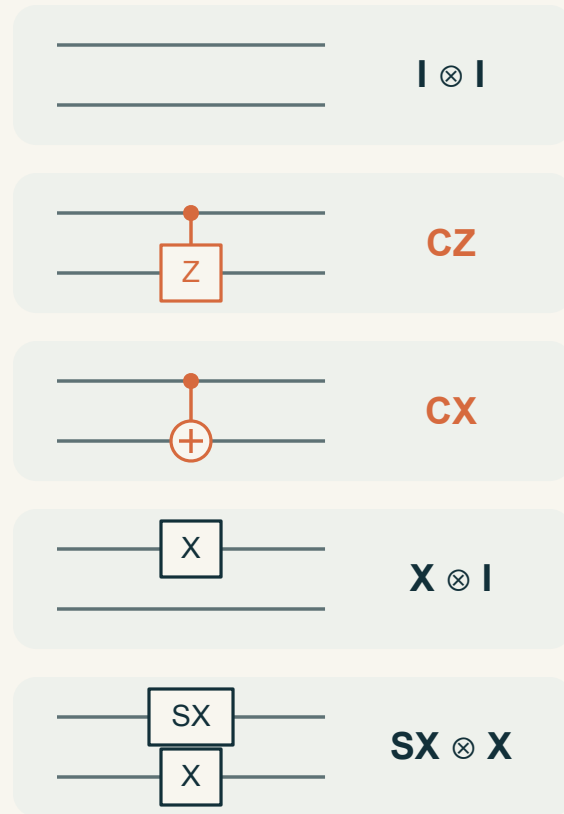
4-dimensional linear space

2 nonzero solution spaces

20

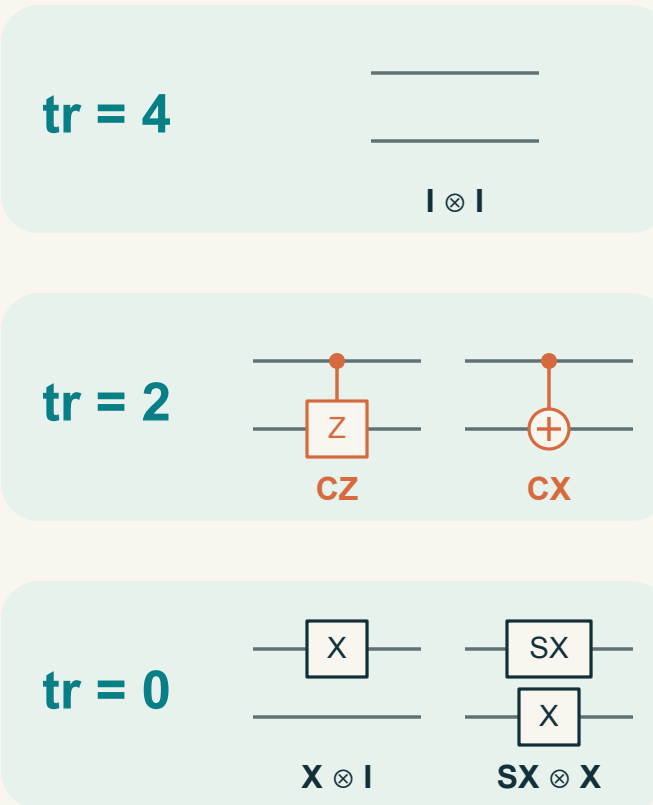
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10-dimensional linear space

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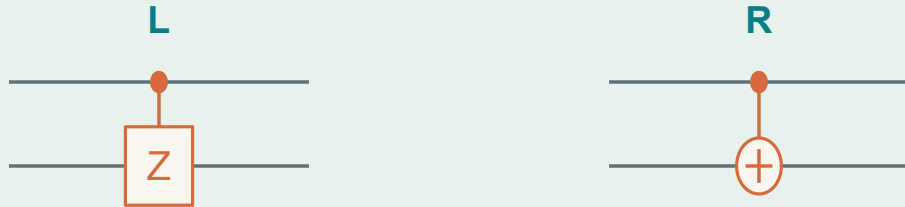
4-dimensional linear space

2 nonzero solution spaces

Eigenvalue test: does a unitary S exist?

4 Compare spectra, including multiplicities

$$(L, R) = (CZ, CX)$$



$$(L, R) = (X \otimes I, SX \otimes X)$$

